

Sequences and Series: Problems, Solutions, and Answers

Problems 14, 17, 24, 25, 31, and 35

Problem 14

Question: The n th term of a sequence is $a_n = n^2 - n$. Find $\sum_{n=1}^{20} a_n$.

Solution:

Substitute the formula for a_n into the sum:

$$\sum_{n=1}^{20} a_n = \sum_{n=1}^{20} (n^2 - n)$$

Separate the two sums:

$$\sum_{n=1}^{20} (n^2 - n) = \sum_{n=1}^{20} n^2 - \sum_{n=1}^{20} n$$

Use the standard formulas

$$\sum_{n=1}^N n = N(N+1)/2, \quad \sum_{n=1}^N n^2 = N(N+1)(2N+1)/6.$$

$$\sum_{n=1}^{20} n^2 = 20 \cdot 21 \cdot 41 / 6 = 2870$$

$$\sum_{n=1}^{20} n = 20 \cdot 21 / 2 = 210$$

$$\sum_{n=1}^{20} a_n = 2870 - 210 = 2660.$$

Answer: 2660

Problem 17

Question: Compute $\sum_{k=1}^{20} 1/[k(k+2)]$.

Solution:

Use partial fractions:

$$1/[k(k+2)] = 1/2 \cdot (1/k - 1/(k+2)).$$

Therefore,

$$\sum_{k=1}^{20} 1/[k(k+2)] = 1/2 \sum_{k=1}^{20} (1/k - 1/(k+2)).$$

Write out the terms. The middle terms cancel:

$$= 1/2 [(1 + 1/2 + 1/3 + \cdots + 1/20) - (1/3 + 1/4 + \cdots + 1/22)]$$

$$= 1/2 (1 + 1/2 - 1/21 - 1/22)$$

$$= 1/2 (3/2 - 43/462) = 1/2 \cdot 650/462 = 325/462.$$

Answer: 325/462

Problem 24

Question: Find the sum $1/2 + 2/4 + 3/8 + 4/16 + 5/32 + \dots + 100/2^{100}$.

Solution:

Let

$$S = \sum_{k=1}^{100} k/2^k = 1/2 + 2/2^2 + 3/2^3 + \dots + 100/2^{100}.$$

Multiply by 2:

$$2S = 1 + 2/2 + 3/2^2 + \dots + 100/2^{99}.$$

Subtract S from 2S. After aligning the terms, we obtain

$$S = 1 + 1/2 + 1/2^2 + \dots + 1/2^{99} - 100/2^{100}.$$

The geometric sum is

$$1 + 1/2 + \dots + 1/2^{99} = [1 - (1/2)^{100}]/(1 - 1/2) = 2 - 2/2^{100}.$$

Thus,

$$S = 2 - 2/2^{100} - 100/2^{100} = 2 - 102/2^{100}.$$

$$S = 2 - 51/2^{99}.$$

Answer: $2 - 102/2^{100}$ (equivalently, $2 - 51/2^{99}$)

Problem 25

Question: Compute $\sum_{k=1}^n k \cdot 2^k$.

Solution:

Let

$$S = \sum_{k=1}^n k \cdot 2^k = 2 + 2 \cdot 2^2 + 3 \cdot 2^3 + \dots + n \cdot 2^n.$$

Multiply by 2:

$$2S = 2^2 + 2 \cdot 2^3 + 3 \cdot 2^4 + \dots + n \cdot 2^{n+1}.$$

Subtract S from 2S:

$$S = -2 - 2^2 - 2^3 - \dots - 2^n + n \cdot 2^{n+1}.$$

Use the geometric-series formula:

$$2 + 2^2 + \dots + 2^n = 2^{n+1} - 2.$$

Therefore,

$$S = -(2^{n+1} - 2) + n \cdot 2^{n+1}$$

$$S = (n - 1)2^{n+1} + 2.$$

Answer: $(n - 1)2^{n+1} + 2$

Problem 31

Question: Compute $1^2/(1 \cdot 3) + 2^2/(3 \cdot 5) + 3^2/(5 \cdot 7) + \dots + n^2/[(2n - 1)(2n + 1)]$.

Solution:

Write the sum as

$$S = \sum_{k=1}^n k^2 / [(2k - 1)(2k + 1)].$$

Since $(2k - 1)(2k + 1) = 4k^2 - 1$, we have

$$k^2 / [(2k - 1)(2k + 1)] = 1/4 + 1/[4(2k - 1)(2k + 1)].$$

Also,

$$1/[(2k - 1)(2k + 1)] = 1/2 [1/(2k - 1) - 1/(2k + 1)].$$

Therefore,

$$S = n/4 + 1/8 \sum_{k=1}^n [1/(2k - 1) - 1/(2k + 1)].$$

The second sum telescopes:

$$S = n/4 + 1/8 (1 - 1/(2n + 1)).$$

$$S = n/4 + n/[4(2n + 1)] = n(n + 1)/[2(2n + 1)].$$

Answer: $n(n + 1) / [2(2n + 1)]$

Problem 35

Question: Find the sum $1/(\sqrt{1} + \sqrt{2}) + 1/(\sqrt{2} + \sqrt{3}) + \dots + 1/(\sqrt{99} + \sqrt{100})$.

Solution:

Rationalize each denominator:

$$\begin{aligned} 1/(\sqrt{k} + \sqrt{k+1}) &\cdot (\sqrt{k+1} - \sqrt{k})/(\sqrt{k+1} - \sqrt{k}) \\ &= [\sqrt{k+1} - \sqrt{k}]/[(k+1) - k] = \sqrt{k+1} - \sqrt{k}. \end{aligned}$$

Thus the sum telescopes:

$$\begin{aligned} (\sqrt{2} - \sqrt{1}) + (\sqrt{3} - \sqrt{2}) + \dots + (\sqrt{100} - \sqrt{99}) \\ = \sqrt{100} - \sqrt{1} = 10 - 1 = 9. \end{aligned}$$

Answer: 9